HDOC Day-65

Tower of Hanoi using Recursion in C

Problem Statement

Write a recursive C program to solve the Tower of Hanoi problem for n disks and display every move. Move all disks from rod A (source) to rod C (destination) using rod B (auxiliary), moving one disk at a time and never placing a larger disk on a smaller disk.

Algorithm

1. Start.
2. Read the number of disks n.
3. If the input is not an integer or n is outside 1 to 10, display an error message and stop.
4. Call towerOfHanoi(n, A, B, C).
5. In towerOfHanoi(n, source, auxiliary, destination), if n equals 1, display the move from source to destination and return.
6. Otherwise recursively move n-1 disks from source to auxiliary, using destination as temporary rod.
7. Display the move of disk n from source to destination.
8. Recursively move n-1 disks from auxiliary to destination, using source as temporary rod.
9. Display the total number of moves, 2^n - 1.
10. Stop.

Test Case Table

Test case	Input (n)	Expected result
1	1	Move disk 1 from A to C; total moves = 1
2	2	A to B (disk 1), A to C (disk 2), B to C (disk 1); total = 3
3	3	7 valid moves; total moves = 7
4	4	15 valid moves; total moves = 15
5	5	31 valid moves; total moves = 31
6	0	Invalid number of disks. Enter 1 to 10.
7	-2	Invalid number of disks. Enter 1 to 10.
8	abc	Invalid input. Enter an integer.
9	10	1023 valid moves; total moves = 1023
10	11	Invalid number of disks. Enter 1 to 10.

C Program

```
#include <stdio.h>

void towerOfHanoi(int n, char source, char auxiliary, char destination)
{
    if (n == 1) {
        printf("Move disk 1 from %c to %c\n", source, destination);
```

```
        return;
    }
    towerOfHanoi(n - 1, source, destination, auxiliary);
    printf("Move disk %d from %c to %c\n", n, source, destination);
    towerOfHanoi(n - 1, auxiliary, source, destination);
}

int main(void)
{
    int n;
    printf("Enter number of disks (1 to 10): ");
    if (scanf("%d", &n) != 1) {
        printf("Invalid input. Enter an integer.\n");
        return 1;
    }
    if (n < 1 || n > 10) {
        printf("Invalid number of disks. Enter 1 to 10.\n");
        return 1;
    }
    printf("Tower of Hanoi moves:\n");
    towerOfHanoi(n, 'A', 'B', 'C');
    printf("Total moves: %d\n", (1 << n) - 1);
    return 0;
}
```

Compilation and Execution Commands

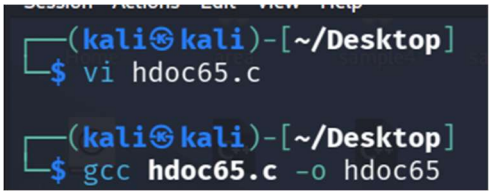
gcc hdoc65.c -o hdoc65

./hdoc65

Result

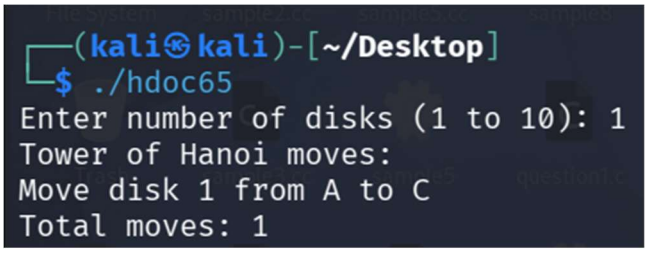
The recursive Tower of Hanoi program displays all required disk moves for valid inputs and reports the correct total number of moves.

Compilation and Execution



Test Case 1

Input: 1 | Expected: Move disk 1 from A to C; total moves = 1



Test Case 2

Input: 2 | Expected: A to B (disk 1), A to C (disk 2), B to C (disk 1); total = 3

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc65
Enter number of disks (1 to 10): 2
Tower of Hanoi moves:
Move disk 1 from A to B
Move disk 2 from A to C
Move disk 1 from B to C
Total moves: 3
```

Test Case 3

Input: 3 | Expected: 7 valid moves; total moves = 7

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc65
Enter number of disks (1 to 10): 3
Tower of Hanoi moves:
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 3 from A to C
Move disk 1 from B to A
Move disk 2 from B to C
Move disk 1 from A to C
Total moves: 7
```

Test Case 4

Input: 4 | Expected: 15 valid moves; total moves = 15

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc65
Enter number of disks (1 to 10): 4
Tower of Hanoi moves:
Move disk 1 from A to B
Move disk 2 from A to C
Move disk 1 from B to C
Move disk 3 from A to B
Move disk 1 from C to A
Move disk 2 from C to B
Move disk 1 from A to B
Move disk 4 from A to C
Move disk 1 from B to C
Move disk 2 from B to A
Move disk 1 from C to A
Move disk 3 from B to C
Move disk 1 from A to B
Move disk 2 from A to C
Move disk 1 from B to C
Total moves: 15
```

Test Case 5

Input: 5 | Expected: 31 valid moves; total moves = 31

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc65
Enter number of disks (1 to 10): 5
Tower of Hanoi moves:
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 1 from A to C
Total moves: 31
```

Test Case 6

Input: 0 | Expected: Invalid number of disks. Enter 1 to 10.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc65
Enter number of disks (1 to 10): 0
Invalid number of disks. Enter 1 to 10.
```

Test Case 7

Input: -2 | Expected: Invalid number of disks. Enter 1 to 10.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc65
Enter number of disks (1 to 10): -2
Invalid number of disks. Enter 1 to 10.
```

Test Case 8

Input: abc | Expected: Invalid input. Enter an integer.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc65
Enter number of disks (1 to 10): abc
Invalid input. Enter an integer.
```

Test Case 9

Input: 10 | Expected: 1023 valid moves; total moves = 1023

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc65
Enter number of disks (1 to 10): 10
Tower of Hanoi moves:
Move disk 1 from A to B
Move disk 2 from A to C
```

```
Move disk 2 from A to C
Move disk 1 from B to C
Total moves: 1023
```

Test Case 10

Input: 11 | Expected: Invalid number of disks. Enter 1 to 10.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc65
Enter number of disks (1 to 10): 11
Invalid number of disks. Enter 1 to 10.
```